NNN-C-688
October 22, 1962

SUPERSEDING
Fed. Spec. GG-C-695a
November 3, 1952
(See section 6)

FEDERAL SPECIFICATION

CRUCIBLE, IGNITION, LABORATORY; METAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers two types of metal ignition crucibles, nickel or platinum, complete with covers, for laboratory use.

1.2 **Classification.**

1.2.1 *Types and capacities.* The crucibles shall be furnished in one or more of the following types or capacities as specified (see 6.1).

Type I—Nickel.
Capacity—100 milliliters (ml.)
250 ml.

Type II—Platinum.
Capacity—20 ml.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following documents, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-T-111—Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inpection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

ASTM Standard:

B162-58T—Specification for Nickel Plate, Sheet, and Strip (Tentative).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Type I crucibles.* Type I crucibles and covers shall be made from 99.4 percent nickel, spun and hardened in conformance with ASTM Designation B162-58T.

3.1.2 *Type II crucibles.* Type II crucibles and covers shall be made from 99.9 percent platinum plus iridium, with the iridium content of not more than 0.4 percent.

3.2 Style and design.

3.2.1 *Crucibles.* Crucibles shall have a standard shape with a flat base to give a firm and stable resting surface. The side wall shall form a sloping and convex surface extending up to the brim. Crucibles shall be of one-piece construction.

3.2.2 *Covers.* Covers shall be flat, sunken in the center, and shall have an extension to serve as a handle. The sunken area in the center shall fit inside the crucible.

3.3 Dimensional requirements.

3.3.1 *Type I crucibles and covers.* Type I crucibles and covers shall conform to the requirements of table I.

3.3.2 *Type II crucibles and covers.*

3.3.2.1 *Type II crucibles.* Type II crucibles shall conform to the requirements of table II.

3.3.2.2 *Type II covers.* Type II covers shall conform to the requirements of table III.

TABLE I. *Dimensions (Type I)*

Crucibles					Covers			
Capacity	Outside* diameter (O.D.)	Bottom* O.D.	Height*	Wall** thickness	O.D.* exclusive of handle	Length* including handle	Inside* diameter of sunken section	Wall** thickness
ml.	mm.	mm.	mm.	mm.	mm.	mm.	mm.	mm.
100	60	30	64	0.625	67	92	58	0.625
250	83	50	83	0.625	92	116	83	0.625

*Tolerance±2 mm.
**Tolerance±0.25 mm.

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TABLE II. *Crucible dimensions (Type II)*

Capacity	Top inside diameter	Bottom inside diameter	Height	Thickness of side	Thickness of bottom	Weight
ml.	mm.	mm.	mm.	mm.	mm.	grams (g.)
20	32.4±0.8	19±1.0	34.5±2.0	0.22±0.05	0.25±0.05	16±1.0

TABLE III. *Cover requirements (Type III)*

Thickness	Outside diameter excluding handle	Length including handle	Diameter of sunken section	Depth of sunken section	Weight
mm.	mm.	mm.	mm.	mm.	g.
0.137±0.01	36.4±0.2	47±0.5	31.5±0.5	3.0±0.3	4.0±0.4

3.4 **Marking.** Crucibles and covers shall be marked with the name or registered trademark of the manufacturer. In addition, type I crucibles and covers shall be marked with letters "Ni" or "N" to indicate that the crucibles are made of nickel.

3.5 **Workmanship.** Crucibles and covers shall be free from defects (scratches, dents, discolorations) or contamination (grease, oil, foreign matter) which detract from their appearance or which impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedures which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1.1, 3.1.2, and section 5 have been satisfied (see 6.1).

4.3 **Lotting.** A lot shall consist of the crucibles and covers of the same type and capacity offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions.

4.4 **Sampling.**

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

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4.5 Examination. Sample crucibles, covers, and level A and level B preparation for delivery requirements shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Crucible and cover.*

Categories	*Defects*
Critical:	None defined
Major:	*AQL 4.0 percent defective*
101	Dimensions incorrect.
102	Defects (scratches, dents, discoloration).
103	Contamination (grease, oil, foreign matter).
104	Marking missing, illegible, or incorrect.

4.5.1.2 *Level A and B preparation for delivery (section 5).*

Categories	*Defects*
	rect, or incomplete.
Critical:	None defined
Major:	*AQL 1.5 percent defective*
101	Unit, intermediate, or shipping container not as specified.
102	Number of crucibles per unit or intermediate container not as specified.
103	Closure of unit, intermediate, or shipping container not as specified.
104	Unit, intermediate, or shipping container marking illegible, incor-

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* One crucible with cover, as specified, constitutes one unit. Each unit shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional; or PPP-B-676, type, class, and style optional. The box shall be provided with interior fittings or sufficient cushioning material to insure against shifting of contents. Box closures shall be secured with tape conforming to UU-T-111.

5.1.1.2 *Intermediate packaging.* The number of unit packages specified (see 6.1), of like type and capacity, crucible and cover, shall be intermediately packaged in a box of appropriate size conforming to type I, class 1, of PPP-B-636. The boxes shall be closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Crucibles shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Crucibles shall be packed in a shipping container conforming to PPP-B-621, class 2; PPP-B-601, overseas type; or PPP-B-636, class 2, W5c or W5s, RSC. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto.

5.2.2 *Level B.* Crucibles shall be packed in accordance with 5.2.1 except that shipping containers shall conform to PPP-B-621, class 1; PPP-B-601, domestic type; or PPP-B-636, type I, class 1, style RSC, fabricated from minimum 275-pound test fiberboard.

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5.2.3 *Level C.* Crucibles shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123, except that the word "platinum" shall not appear on the shipping container.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129, except that the word "platinum" shall not appear on the shipping container.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packaging and packing required.

(c) Type and capacity of crucible required.

(d) Number of unit packages per intermediate packages required.

6.2 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard and military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.3 **Supersession data.** This specification includes the requirements of Military Specification MIL-C-3731, dated 11 April 1952.

6.4 **Transportation description.** Transportation description applicable to this item is:

Type I.—Nickel.
Crucibles, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.
Crucibles, not otherwise indexed.
Motor volume minimum weight 24,000 pounds.
Type II.—Platinum.
Not taken by freight. Ship via Express or parcel post.

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